YIN SAN GARY KWAN

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Work Experience

Freelance Software Engineer *Jan 2026 – Aug 2026*

Self-employed

- Developed a computer vision auto-framing pipeline for sports highlight videos with object detection, player tracking, movement analysis, and automated cropping for stable social-media-ready clip generation.
- Built a Player Album system for sports highlight videos with segmentation, multi-object tracking, person re-identification embeddings, clustering, conflict validation, and identity validation for recurring player grouping across clips.
- Built a Text Insight RAG system for local text understanding, converting cleaned document chunks into LangChain documents for semantic search, sentiment analysis, HuggingFace embedding retrieval, lexical rescue, reranking, and Ollama-based English answers.

Research Assistant *Jun 2025 – Dec 2025*

Department of Electronic Engineering, The Chinese University of Hong Kong

- Assisted in daily research activities at the Digital Signal Processing and Speech Technology Laboratory.
- Conducted research on Cantonese speech datasets for evaluating and improving text-to-speech AI models.
- Performed data preprocessing and data cleaning on Cantonese speech audio and corresponding transcripts.
- Trained speaker-specific text-to-speech models based on a guided diffusion model architecture.
- Enhanced clients' pre-recorded audio to improve the performance of their text-to-speech models.
- Improved the understandability, portability, and user-friendliness of the model training and inferencing codebase.

Software Engineer Intern *Jul 2024 – Aug 2024*

HairCoSys Limited

- Applied data preprocessing techniques to raw hair follicle images to enhance model performance.
- Trained object detection models based on the YOLO framework and object classification models using the MMClassification framework to identify and differentiate various hair follicle diseases.
- Developed and integrated prediction and post-processing functionalities into the AI-driven haircare application.

Software Engineer Intern *Jun 2023 – Aug 2023*

Intelligent Design Technology

- Utilized data preprocessing techniques on raw thermal images to improve model performance.
- Developed and trained an object detection model based on the YOLO framework for identifying mice using thermal camera data.

Education

Bachelor of Engineering in Electronic Engineering *Aug 2020 – Jul 2025*

Minor in Computer Science, Information Engineering, Web and Cloud Computing

The Chinese University of Hong Kong (CUHK)

Thesis title of Final Year Project:

A Deep Learning Re-Implementation for Monocular 3D Object Detection with Complementary Depths in MMDetection

- Re-implemented the MonoCD framework in MMDetection to improve monocular 3D object detection using complementary depth estimation.
- Designed a modular 3D reconstruction pipeline that fuses 2D object detections with learned depth maps to estimate 3D bounding boxes on the KITTI dataset.

Associate of Engineering *Aug 2018 – Jul 2020*

The University of Hong Kong – School of Professional and Continuing Education (HKU SPACE) Community College

Computer Science Theme, Principal's Honours List in year 2019 – 2020

Skills

Artificial Intelligence & Machine Learning:

Artificial Intelligence, Machine Learning, Deep Learning, Pattern Recognition, Model Training, Model Evaluation, Model Optimisation, Error Analysis

Computer Vision & Video Processing:

Computer Vision, Object Detection, Object Classification, Object Segmentation, Player Tracking, Ball Tracking, Person Re-Identification, Video Processing, Auto-Framing, Crop Generation, Camera Calibration, Monocular 3D Object Detection, Depth Estimation

Natural Language Processing, RAG & Speech AI:

Natural Language Processing, Retrieval-Augmented Generation, Semantic Search, Sentiment Analysis, Text Chunking, LangChain Retrieval, HuggingFace Embeddings, Sentence Transformers, Ollama-Based Local LLM Inference, Text-to-Speech,

Speech Processing, Audio Processing

Data Processing & Retrieval:
Data Analytics, Data Analysis, Data Visualization, Data Preprocessing, Data Cleaning, Dataset Organisation, Vector Retrieval, Cosine Similarity, Embedding-Based Clustering, Agglomerative Clustering, Lexical Rescue, Reranking

Programming Languages:
Python, SQL, C++, C, Java, JavaScript, Dart

Frameworks & Libraries:
PyTorch, OpenCV, MMClassification, MMDetection, YOLO, YOLOE, Ultralytics, Torchreid, OSNet-AIN, LangChain, Hugging Face, OpenGL, Flutter

Software & Application Development:
Software Development, Software Design, Data Structures, Web Development, Web Application Development, Mobile Application Development, Command-Line Application Development, Debugging, Testing, Documentation

Tools & Platforms:
Git, GitHub, Linux, Conda, FFmpeg, CUDA, JSON, Markdown, KITTI Dataset, COCO Annotation Format

Robotics:
Robotics, Robot Programming

Project & Management Skills:
Project Management, Project Planning, Modular Pipeline Design, Technical Documentation

Leadership

External Vice President *Feb 2022 – Jan 2023*
Electronic Engineering Society, CUHK
- Assisted the President in all duties related to external affairs.
- Coordinated communication with other student organizations.

Orientation Camp Organizer *May 2021 – Sep 2024*
United College, CUHK
- Drafted, scheduled, and organized orientation camps for freshmen of the United College, CUHK.